

Ștoc Iustin

CONTACT ME

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Github

PROFİLE

Automation engineering student passionate about software development and control systems. I'm developed in both fields: IT and automation, with a high capacity to increase my knowledge, skill of communication and team working and being involved in projects.

EDUCATİON

Bachelor's Degree in Automation & Computer Science: Technical University of Cluj-Napoca 2022-present

High school Diploma in Math and Informatics: National College "Emanuil Gojdu" 2017-2021

EXPERIENCES

→ **summer of 2024**
Waiter/Rod 'N' Reel Resort
-It was a summer experience with Work and Travel program in United States, and it was a pleasant time where *I achieved a better communication skill*.

→ **2021-2022**
-SMT Operator/Plexus
This job was more significant for the field I'm studying, only because I chose to study in Cluj, I had to give it up because it was more profitable to work as a waiter only on weekends and on vacation to be able to study at the university, this position proved to me that *I know how to adapt and learn new information quickly*, which is why in the end before I left I was offered the position of technician.

→ **2019 to 2024**
-Waiter/Mediterana Ristorante
I was a waiter in this restaurant with an experience of 5 years in this field, I worked while I was in high school and college on free weekends and holidays, I *undertook the organization of events and the organization of the staff* in the restaurant.

SKİLLS

→ Design and analysis of control systems(Matlab) **9/10**
→ PLC programming(Siemens TIA Portal) **7/10**
→ Embedded system and microcontrollers(STM32) **8/10**
→ Python(Image processing) **8/10**
→ C/C++ **9/10**
→ Java **8/10**
→ C# **9/10**

PROJECTS

1)Personalized Anesthesia Control<->ongoing
→ Currently working on a scientific paper focused on the design and implementation of a fractional-order supervisory controller for multi-input general anesthesia. This project combines MATLAB-based modeling with embedded system concepts and aims to improve precision and adaptability in drug delivery.

LANGUAGES

English

French

2)ProMed<->ongoing
→A full-stack web platform developed using ASP.NET, Angular, and SQL Server for managing public hospital appointments. Includes doctor-patient scheduling, secure login, and administrative dashboard. Demonstrates capabilities in software architecture, REST API development, and database integration.

3)AI-based Traffic Optimization
→Developed a smart traffic control solution using YOLOv8 for object detection on real-time video feeds. Implemented traffic light control logic based on detected congestion patterns. Used Python and OpenCV, integrating image processing with automation control strategies.

4)Biomedical Signal Processing – ECG with STM32
→Designed the software architecture for an ECG signal acquisition and processing system targeting STM32 microcontrollers. The system simulates analog front-end (AFE) signal capture and applies digital filtering (low-pass, notch filters) using CMSIS-DSP library. The output includes heartbeat detection and display of filtered signal over UART. Currently developed in a simulated environment with plans for future implementation on STM32 boards.

5)Smart Energy Meter Monitor with STM32
→Designed a system for real-time monitoring and logging of electrical energy consumption using STM32 architecture. The project simulates reading analog voltage and current inputs, calculates active power using RMS methods, and logs the data over UART or SD card interface. The design includes overcurrent protection logic, power factor estimation, and live display on a 16x2 LCD or via serial monitor.

6)Ultrasonic Radar System with Arduino
→Designed and implemented an ultrasonic radar system using an Arduino Uno, an HC-SR04 ultrasonic sensor, and a SG90 servo motor. The system performs area scanning by rotating the sensor across 180°, capturing real-time distance data and mapping obstacles. Results were displayed via serial interface and optionally visualized on an LCD.